School of Information Technology, Engineering, Mathematics and Physics

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CS230: Requirements Engineering
Semester 1, 2023
Mode: Face to Face

Short Test 2
Date: Friday 17 May 2023, Time: 11.00 – 12.50am (FJT)

Time Allowed: 50mins
30 marks (Weight: 7.5%)

Student ID: _____
Student Name: _____

LEARNING OUTCOMES:

- Identify various requirement elicitation techniques for software development.

INSTRUCTIONS

1) This exam has two sections:
   a) Section A: Short-answer questions (20 marks)
   b) Section B: Case Study (10 marks)
2) Answer all questions in this booklet. All questions are compulsory.
3) The total number of pages including this cover sheet is 6. Extra answer sheet is also attached.

~ALL THE BEST~

Section A: Short Answer Questions *(15 Marks)*

Question 1 (5 marks)

Answer the question based on the system requirements specification.

a) What does Abstraction and Modeling, Decomposition and Hierarchy solve? (1 mark)
complexity / Complexity

b) Write the type of hierarchy is this "University organizational structure: the VC is superior to the Head of Schools, who are superior to their Discipline Coordinators who are superior to their teaching staff". (1 mark)
Part-of

c) Difference between Inception and Elaboration? (1 mark)

Elaboration focuses on requirements capture and determining the structure of the system. Inception is concerned with determining the scope and purpose of the project.

d) List and explain two Characteristics of a Good SRS. (2 marks)

1. Unambiguous
2. Complete
3. Verifiable
4. Consistent
5. Modifiable
6. Traceable
7. Usable during the Operation and Maintenance Phase

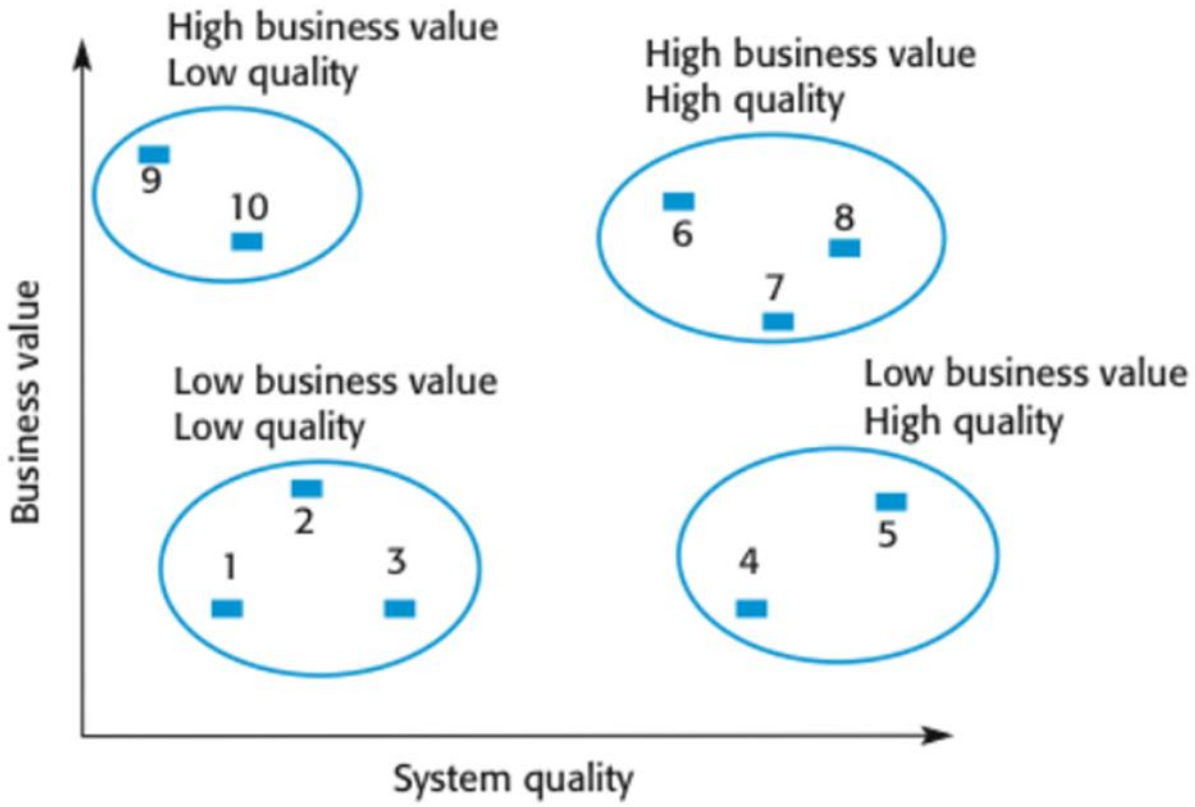
Question 2 (5 marks)

Answer the question based on the software evolution.

a) Bula IT developed new software Bula Hangout and does continuous changes over time. Give any two reasons why software changes are inevitable. (1 mark)

- New requirements emerge when the software is used;
- The business environment changes;
- Errors must be repaired;
- New computers and equipment is added to the system;
- The performance or reliability of the system may have to be improved.

b) Organizations that rely on legacy systems must choose a strategy for evolving these systems. The chosen strategy should depend on the system quality and its business value. Refer to system quality and its business value given below and explain the four outputs while dealing with an organization? (4 marks)



Solution:

- Low quality, low business value: **should be scrapped.**
- Low-quality, high-business value: **make an important business contribution but are expensive to maintain. Should be re-engineered or replaced if a suitable system is available.**
- High-quality, low-business value: **replace with COTS, scrap completely, or maintain.**
- High-quality, high business value: **continue in operation using normal system maintenance.**

Mark Allocation: four correctly answered = full mark. Partial or correct ones get 1 mark each

Question 3 (5 marks)

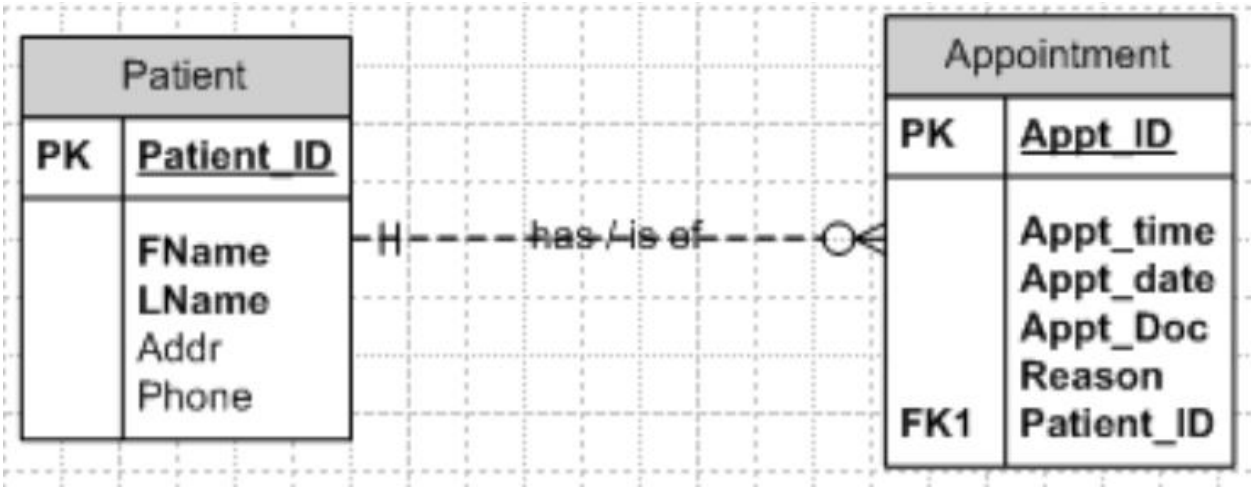
a) Program evolution dynamics is the study of the processes of system change. After major empirical studies, Lehman and Belady proposed that there were a number of 'laws' which applied to all systems as they evolved.

List and describe any three Lehman's law. (3 marks)

Law	Description
Continuing change	A program that is used in a real-world environment must necessarily change, or else become progressively less useful in that environment.
Increasing complexity	As an evolving program changes, its structure tends to become more complex. Extra resources must be devoted to preserving and simplifying the structure.
Large program evolution	Program evolution is a self-regulating process. System attributes such as size, time between releases, and the number of reported errors is approximately invariant for each system release.
Organizational stability	Over a program's lifetime, its rate of development is approximately constant and independent of the resources devoted to system development.

Law	Description
Conservation of familiarity	Over the lifetime of a system, the incremental change in each release is approximately constant.
Continuing growth	The functionality offered by systems has to continually increase to maintain user satisfaction.
Declining quality	The quality of systems will decline unless they are modified to reflect changes in their operational environment.
Feedback system	Evolution processes incorporate multiagent, multiloop feedback systems and you have to treat them as feedback systems to achieve significant product improvement.

b) Find below a diagram representing the classes with their associations for a medical healthcare appointment system. Explain the associations between both patient and appointment (2 marks)



- Any one patient (Bill Jones, for example) may have no appointments but may have many.
- Any one appointment (#2191 at 4 p.m. on 5/21) must be for one patient and no more than one

Mark Allocation: both correctly answered = full mark. Partial or correct ones get 1 mark each

Section B: Case Study *(15 Marks)*

Vuvale Hypermarket Dispatch System

The purpose of the Vuvale Dispatch System is to provide ordering and delivery logistics for the supermarket.

Initially, prospective customers place an order request for selected items. After order is received, the salesperson checks the inventory for the availability of items on the order. To verify this, the salesperson has to contact the dispatch clerk in the warehouse. If there is not enough stock, the salesperson cancels the order. However, if there are enough stock to meet the order, the salesperson then requests the credit card information and at the same time the shipping address from the customer. The shipping details are then passed on to the dispatch clerk at the warehouse who then packs the order and ships it to the address.

For delivery purposes, each shipment is assigned to a delivery person. The delivery person is responsible to deliver the package to the given address.

a) Identify and List the use cases with their brief descriptions. (5 marks)

Actors:

Buyer/Customer, Sales, Warehouse (1.5 marks)

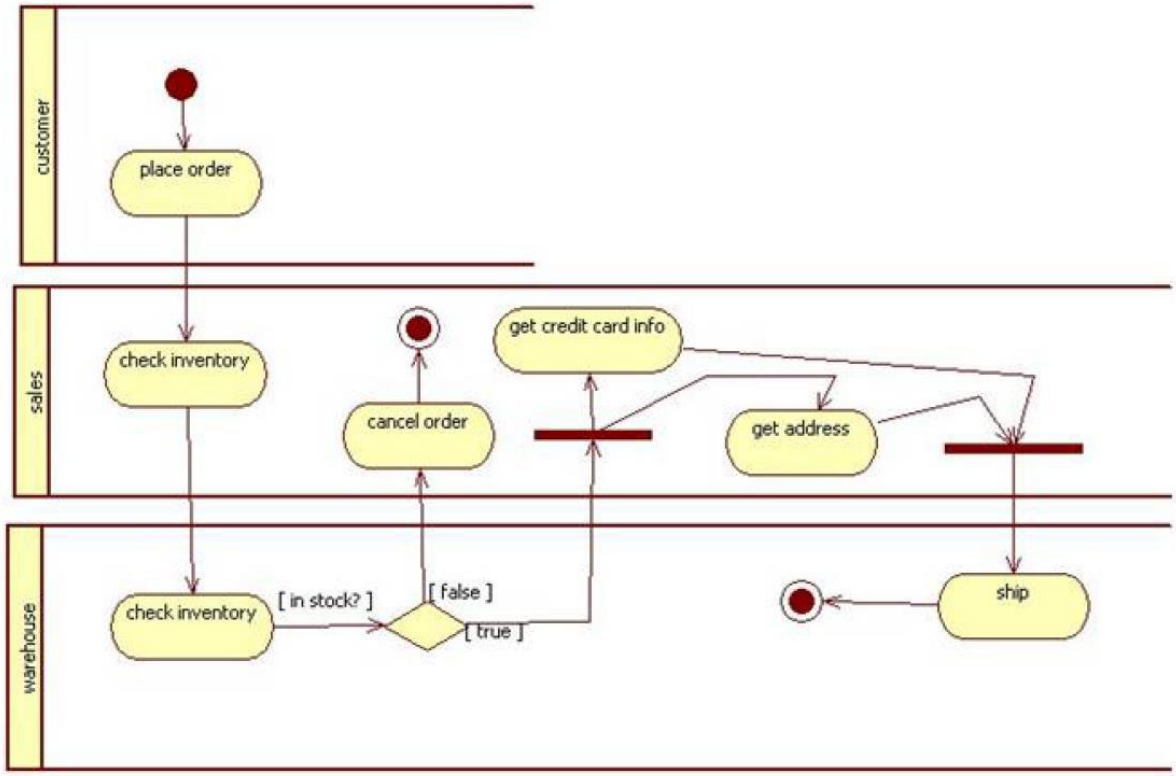
Use Cases

User	User goal & Resulting Use Case

Buyer	Place order
Sales	Check inventory Cancel order Request credit card info and address
Warehouse	Check inventory Inform sales about stock Ship the order

- ½ mark each use case. (3.5 marks)

b) Develop an Activity diagram based on the above case study. Note any ambiguities or questions that you have as you develop the model. If you need to make assumptions, also note them. (10 marks)



Note: diagram can be portrait oriented too, way discussed in the lecture and lab

Mark Allocation:

- Identified all activities correctly (3 marks)
- All the data flow correctly (3 marks)
- Correct decision flow (3 mark)
- Correct activity diagram structure and boxes (1 mark)

THE END

EXTRA ANSWER SHEET – USE ONLY WHEN NEEDED

- Mark the question number properly if used.